Annex E

Epistemic Computational Units (ECU / UCE) Formal Specification and Design Patterns

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Abstract

This annex provides a formal specification of *Epistemic Computational Units* (ECUs), also referred to as *Unità Computazionali Epistemiche* (UCEs) in the Italian texts, as they are used in the RLA/CRC and ECNN framework. ECUs are designed as the core epistemic elements of epistemic neural architectures: they receive high–level representations from a representation channel, interpret them as structured claims in a given domain, and produce *epistemic artifacts* comprising decisions, confidence, explicit "unknown" and "contradiction" signals, and human–readable rationales. Their behaviour is constrained by a deterministic *epistemic matrix* M, encoding entities, rules, coherence constraints and rejection policies, so that the overall system remains Turing–computable and internally falsifiable [Turing, 1936/1937, Shalizi & Crutchfield, 2001].

At a higher tier, small but structured reticula of ECUs/UCEs implement higher–order epistemic functions such as executive control, valuation, memory consolidation and metacognitive monitoring. They instantiate, at the highest levels of an ECNN, compact reticula in the sense of Annex A, where the basic "neurons" are themselves epistemic units.

ECUs/UCEs can be realised in different regimes, ranging from purely symbolic rule–based systems to orchestrated assemblies of Large Language Models (LLMs) acting as "epistemic neurons" under strictly controlled prompts and parsing protocols [Brown et al., 2020, Ouyang et al., 2022, Wei et al., 2022]. The annex covers: (i) concepts and definitions; (ii) architectures for minimal and LLM–based ECUs; (iii) multi–ECU topologies and their role in compact reticula; (iv) determinism, computability and epistemic closure; (v) inference and self–audit protocols; (vi) metrics and test suites for epistemic performance; (vii) design patterns and deployment scenarios. Throughout, ECUs/UCEs are presented as *epistemic* devices for structuring and exposing knowledge and ignorance, not as ontological models of cognition.

Contents

1 Introduction and Scope

Deep neural networks excel at pattern recognition but typically offer limited support for explicit notions of explanation, contradiction, or admitted ignorance [Lipton, 2018, Miller, 2019]. Standard architectures are optimised for predictive accuracy; their logical limits remain implicit in calibration errors, adversarial fragility and opaque failure modes.

Epistemic architectures extend this picture. They incorporate:

- a *representation channel* (e.g. a convolutional, recurrent or transformer backbone) that compresses raw data into abstract features;

- an *epistemic head* that issues not only labels and probabilities but also explicit "unknown" and "contradiction" statements;
- an internal *epistemic operator* that evaluates candidate decisions against a domain theory and explicit epistemic policies.

Epistemic Computational Units (ECUs/UCEs) are the atomic components of this operator. They are intended as:

1. finite, deterministic devices constrained by a domain–specific epistemic matrix M;
2. capable of converting high–level representations into structured artifacts containing decisions, rationales and epistemic signals;
3. auditable objects whose behaviour can be stress–tested by challenge suites and falsifiability protocols.

At a higher tier, small multi–ECU reticula with specialised roles and fixed topology provide an interface between ECNN representation channels (Annex C) and the multi–level reticular semantics of Annex A and Annex D.

1.1 Relation to RLA/CRC and ECNNs

In the language of Reticular Local Abstraction (RLA) and Compact Reticular Computability (CRC) from Annex A, ECUs/UCEs and their reticula play three roles:

- Each ECU/UCE is a Turing–computable map between a representation level L_z (e.g. the penultimate layer of an ECNN) and an epistemic level L_{ep} carrying artifacts.
- A multi–ECU reticulum is a compact RLA sub–reticulum comprised of such levels and transmissions, often with additional internal state (context buffers, episodic memory).
- ECNNs of Annex C are extended, at their top, by such reticula so that epistemic behaviour (unknowns, contradictions, rationales) becomes an explicit part of the compact reticulum rather than an external post–hoc diagnostic.

This annex concentrates on the local design of ECUs/UCEs and their small reticula, assuming the global reticular context from Annex A and the ECNN layer structures of Annex C.

1.2 Epistemic, not ontological, stance

The treatment is explicitly epistemic. ECUs/UCEs encode and expose structured knowledge and ignorance about a domain relative to a given epistemic horizon and theory, but do not claim to implement any privileged ontology of human or biological cognition [Anderson, 1972, Noble, 2012]. Their value lies in:

- making limits and contradictions visible as first–class objects;
- enabling systematic falsification attempts [Popper, 1959];
- providing an interface between learned representations and symbolic domain theories, in the spirit of neural–symbolic integration [Besold et al., 2017, d'Avila Garcez et al., 2012].

2 Concepts and Basic Definitions

2.1 Epistemic entities and artifacts

Definition 2.1 (Epistemic entities)**.** *An* epistemic entity *is any concept, predicate or label used in a given domain theory. We write $E = \{e_1, \dots, e_m\}$ for the finite set of entities, such as diagnostic labels, risk categories, legal norms, causal regimes, or physical phases.*

Definition 2.2 (Epistemic artifact)**.** *An* epistemic artifact *is a finite structured object*

$$a = (y^\star, q, s, R, \Pi),$$

where:

- $y^\star \in E \cup \{\bot, \bot_c\}$ *is a primary decision, with* $\bot$ *denoting "unknown" and* $\bot_c$ *denoting "contradiction";*
- $q \in [0, 1]$ *is a confidence or epistemic strength parameter;*
- s *is an optional signature or hash referencing the input and internal state (for audit and reproducibility);*
- R *is a finite set of rationales (structured or textual), e.g. tuples of entities and rules supporting* $y^\star$*;*
- Π *is a finite set of policies or actions recommended under the given decision.*

Artifacts are the principal interface between ECUs/UCEs and downstream systems or human users. They can be seen as compressed summaries of the epistemic stance of the system on a particular input.

2.2 Epistemic matrix

Definition 2.3 (Epistemic matrix)**.** *An* epistemic matrix *is a tuple*

$$M = (E, R, C, P),$$

where:

- E *is a finite set of epistemic entities;*
- R *is a finite set of inference rules between entities (implications, equivalences, defaults, temporal rules);*
- C *is a finite set of coherence constraints, encoding incompatibilities, subsumption and other structural relations (e.g.* e_i *and* e_j *cannot both hold, or one subsumes the other);*
- P *is a finite set of policies specifying conditions for rejection, abstention, escalation, or fallback actions.*

All components of M are represented in data structures suitable for processing by a Turing machine [Turing, 1936/1937].

The epistemic matrix plays three roles:

1. it constrains which primary decisions $y^\star$ are allowed in which contexts;
2. it defines what counts as contradiction or incoherence;
3. it codifies domain–specific prudence (when to abstain, escalate, or override).

2.3 Epistemic Computational Unit (ECU / UCE)

Definition 2.4 (Epistemic Computational Unit)**.** *Given a representation space Z (e.g. $\mathbb{R}^d$) and an epistemic matrix M, an* Epistemic Computational Unit *(ECU) is a deterministic, Turing–computable function*

$$\mathrm{ECU}_\phi : Z \times M \longrightarrow \mathcal{A},$$

where $\mathcal{A}$ is a finite or countable set of epistemic artifacts and ϕ denotes internal configuration (weights, thresholds, rule parameters).

Definition 2.5 (UCE notation)**.** *In the Italian texts the acronym UCE (*Unità Computazionale Epistemica*) is used as a synonym for ECU. No formal distinction is made in this annex; the choice of notation reflects linguistic context rather than mathematical content.*

An ECU/UCE is thus a local reticular map from a representation level $L_z = \langle Z, \Sigma_z \rangle$ (as in Annex C) to an epistemic level $L_{\mathrm{ep}} = \langle \mathcal{A}, \Sigma_{\mathrm{ep}} \rangle$, subject to the constraints of M.

2.4 Multi–ECU reticula

Definition 2.6 (Multi–ECU reticulum)**.** *A* multi–ECU reticulum *is a finite RLA sub–reticulum*

$$\mathcal{U} = (L_z, L_{\mathrm{ep}}^{(1)}, \dots, L_{\mathrm{ep}}^{(K)}, L_{\mathrm{glob}}),$$

where:

- *L_z is a representation level receiving inputs from an ECNN or another model;*
- *each $L_{\mathrm{ep}}^{(k)}$ is the range of an ECU/UCE $\mathrm{ECU}^{(k)}$, capturing the epistemic stance of a specialised unit (e.g. legal, risk, causal);*
- *L_{glob} is a global aggregation level that fuses the artifacts from all $L_{\mathrm{ep}}^{(k)}$ into a joint artifact a_{glob}.*

Transmissions within $\mathcal{U}$ are:

- *$\tau_z^{(k)} : Z \to \mathcal{A}^{(k)}$, $(z \mapsto \mathrm{ECU}^{(k)}(z, M))$;*
- *$\tau_{\mathrm{agg}} : \mathcal{A}^{(1)} \times \cdots \times \mathcal{A}^{(K)} \to \mathcal{A}_{\mathrm{glob}}$, a deterministic aggregation map.*

Informally, ECUs/UCEs are the atomic epistemic units; multi–ECU reticula such as $\mathcal{U}$ are the structured "organs" made of ECUs/UCEs, responsible for a coherent epistemic interface at the top of an ECNN or other reticulum [Besold et al., 2017, d'Avila Garcez et al., 2012].

3 Architectures for ECUs

3.1 Minimal (non–LLM) ECUs

Pipeline. A minimal ECU follows the pipeline:

$$z \in Z \xrightarrow{\Psi} h \in H \xrightarrow{\Gamma_M} \hat{a} \in \mathcal{A} \xrightarrow{\Pi_M} a \in \mathcal{A},$$

where:

- Ψ is a neural or linear projection from Z to a symbolic hypothesis space H (e.g. logits over E, or scores over rule premises);
- Γ_M applies rules and constraints from M to h, producing an intermediate artifact $\hat{a}$;
- Π_M enforces policies such as abstention, escalation or default actions, yielding the final artifact a.

Example construction. Let $H = \mathbb{R}^{|E|}$ be a logit vector h over entities, with h_k proportional to the logit for entity e_k. Let $r_{ij} \in R$ denote a rule $e_i \Rightarrow e_j$. A simple deterministic update is:

$$h_j \leftarrow \max(h_j, h_i - \delta_{ij}),$$

for some fixed penalty $\delta_{ij} \geq 0$. Constraints in C (e.g. e_i incompatible with e_j) are enforced by setting the lower–scored entity to $-\infty$. Policies in P then map the resulting h to:

- a decision $y^\star = \arg\max_k h_k$, if coherence holds and confidence exceeds a threshold;
- $y^\star = \perp$ if confidence is low or rules prescribe abstention;
- $y^\star = \perp_c$ if mutually incompatible entities are simultaneously activated.

Rationales R are constructed from the subset of rules and entities that contributed to the final decision; policies Π are derived from P.

3.2 LLM–based "epistemic neurons"

Motivation. For complex domains, the hypothesis space H may be difficult to encode with simple logits and rules alone. Large Language Models (LLMs) offer a practical way to approximate rich inference patterns, but they must be constrained so that the overall ECU/UCE remains deterministic, Turing–computable and auditable [Brown et al., 2020, Ouyang et al., 2022, Wei et al., 2022].

Definition 3.1 (Epistemic neuron). *An* epistemic neuron *is an LLM instance with parameters ω, executed under:*

1. *a fixed prompt template that encodes M or a subset M_k;*
2. *a fixed decoding policy (e.g. temperature $T = 0$, top–$k = 1$);*
3. *a strict output format (e.g. JSON schema) enforced by a deterministic validator.*

The neuron receives a structured representation of z as input and emits a candidate partial artifact $a^{(k)}$.

Multi–neuron ECUs. An ECU/UCE may aggregate the outputs of K epistemic neurons:

$$z \longmapsto (a^{(1)}, \dots, a^{(K)}) \longmapsto a,$$

where the aggregation is entirely deterministic. For example, each neuron can be specialised on a subdomain (e.g. legal norms, risk metrics, causal explanations). An aggregation policy may:

- detect contradictions among $a^{(k)}$ and map them to $\perp_c$;
- compute a consensus decision and confidence from the $y^{\star(k)}$ and $q^{(k)}$;
- compose rationales into a single structured explanation, while enforcing the coherence constraints C.

Determinism and computability. Despite using LLMs, the overall ECU/UCE remains Turing–computable when:

- prompts, decoding parameters and maximum output length are fixed and documented;
- outputs are post–processed by a deterministic parser and validator;
- any stochastic components are either disabled or simulated in a way that can be reproduced (e.g. logged seeds).

Under these conditions, the LLMs act as large but finite transition tables, not as non–computable oracles [Turing, 1936/1937].

3.3 Multi–ECU topologies

At the level of multi–ECU reticula, ECUs/UCEs are connected in small reticula that implement distinct epistemic functions:

- **Executive ring:** a cycle of ECUs/UCEs handling goal management, constraint enforcement and conflict resolution over artifacts.
- **Valuation ring:** ECUs/UCEs specialised in assessing risk, cost, utility, or other valuation metrics, feeding into policies Π.
- **Memory ring:** ECUs/UCEs equipped with access to episodic or semantic stores, controlling how past artifacts influence current decisions.
- **Metacognitive ring:** ECUs/UCEs operating on artifacts produced by other ECUs/UCEs, checking coherence, stability and calibration.

These rings can themselves be organised in higher–level reticula, provided IOA, EC and TC (Annex A) remain satisfied.

4 Determinism, Computability and Epistemic Closure

4.1 Deterministic ECU regimes

Axiom 4.1 (Deterministic execution)**.** *An ECU/UCE must be executable as a deterministic program: for each pair* (z, M) *it returns the same artifact* a*, up to representational equivalence, across runs.*

This axiom excludes untracked randomness inside the ECU/UCE. Stochasticity can still be simulated (e.g. for exploration), but it must be made explicit and reproducible so that auditability is preserved.

4.2 Epistemic closure for ECUs and multi–ECU reticula

Definition 4.1 (Epistemic closure for ECUs and multi–ECU reticula)**.** *An ECU/UCE or a multi–ECU reticulum is* epistemically closed *with respect to* M *if all its decisions, rationales and policies can be expressed as finite statements involving only:*

- *entities in* E*;*
- *rules in* R*;*
- *constraints in* C*;*
- *policies in* P*;*
- *deterministic functions over the representation space* Z *and internal buffers.*

Intuitively, closure ensures that ECUs/UCEs and their reticula do not silently rely on unstated assumptions or external black boxes that cannot be mapped back into M. In practice, this calls for documentation artefacts such as:

- JSON schemas describing artifact formats;
- formal descriptions (or at least structured encodings) of rule sets and constraints;
- versioned prompt templates for LLM–based epistemic neurons.

4.3 Internal epistemic "oracle" (non–metaphysical)

ECUs/UCEs are sometimes colloquially described as providing an "internal epistemic oracle": a capability to say *I do not know* or *this is contradictory* from within the model. Formally, this is realised by:

1. explicit $\perp$ and $\perp_c$ states inside artifacts;
2. deterministic policies in P that map detectable conflict or lack of justification to those states;
3. challenge suites used to test that the ECU/UCE uses $\perp$ and $\perp_c$ in the intended way.

No non–computable oracle is implied; the term is purely epistemic and relative to the horizon specified by M [Popper, 1959].

5 Inference and Self–Audit Protocols

5.1 Forward inference

A generic ECU/UCE inference protocol takes as input an internal representation z and returns an artifact a:

1. **Claim extraction:** apply a deterministic map $\Phi : Z \to H$ that interprets z as candidate claims (e.g. scores over entities, structured features).

2. **Rule application:** apply Γ_M to obtain an intermediate artifact $\hat{a}$, enforcing inference rules and coherence constraints.

3. **Policy application:** apply Π_M to $\hat{a}$, producing the final artifact a with decisions and policies.

4. **Formatting:** render a into the agreed external schema (e.g. JSON with standard fields).

In LLM–based ECUs/UCEs, step 1 may itself call one or more epistemic neurons whose outputs are then consolidated by Γ_M and Π_M.

5.2 Self–audit and stability checks

To make logical limits explicit, ECUs/UCEs and their reticula can implement self–audit protocols:

- **Perturbation stability:** apply controlled perturbations δz to inputs or intermediate representations and check whether decisions and rationales remain stable within expected ranges.

- **Consistency checks:** re–apply inference with subsets of rules or altered policies to detect internal brittleness or hidden dependencies.

- **Round–trip sanity:** given an artifact a, project it back to a synthetic representation z', feed z' to the ECU/UCE and verify that the resulting artifact a' is compatible with a.

These procedures allow ECUs/UCEs and their reticula to approximate a form of meta–reasoning about their own outputs, without departing from Turing–computability.

5.3 Interaction with Popper–χ suites

ECUs/UCEs and their reticula are naturally tested using Popper–style challenge suites as described in Annex C and Annex D:

- cases engineered to expose inconsistencies in M;

- corner cases where domain theory is genuinely agnostic;

- adversarial constructions that mimic in–domain statistics while reversing or scrambling semantics.

The ECU/UCE's role is to respond with $\perp$ or $\perp_c$ whenever M cannot support a determinate decision, turning model limits into observable behaviours [Popper, 1959, Shalizi & Crutchfield, 2001].

6 Metrics and Test Suites for ECUs and Reticula

6.1 Single–ECU metrics

Let $\mathcal{D} = \{(z_i, y_i, e_i)\}_{i=1}^{N}$ be a dataset where z_i are representations, y_i labels or ground–truth epistemic entities, and e_i indicate epistemic regime (*certain*, *uncertain*, *contradictory*).

Core metrics include:

- **Accuracy on certain cases:** standard classification accuracy on the subset $\{i : e_i = \text{certain}\}$.
- **Abstention quality:** the fraction of cases with $e_i = \text{uncertain}$ for which the ECU/UCE outputs $\perp$.
- **Contradiction detection:** the fraction of contradictory cases correctly flagged as $\perp_c$.
- **Calibration:** expected calibration error (ECE) for the confidence scores q_i in artifacts [Guo et al., 2017].
- **Explanation coherence:** the proportion of explanations in R that satisfy the constraints C (e.g. do not cite mutually incompatible entities).

6.2 Multi–ECU reticulum metrics

In architectures with multiple ECUs/UCEs inside a reticulum, additional metrics become relevant:

- **Epistemic diversity:** the average pairwise distance between artifacts produced by different ECUs/UCEs on the same inputs, measured over rationales or decision structures.
- **Consensus strength:** the fraction of cases where ECUs/UCEs agree on $y^\star$ and q, indicating stable conclusions.
- **Conflict resolution quality:** in cases of disagreement, how often the aggregation policy yields decisions aligned with ground–truth or expert judgement.
- **Metacognitive reliability:** for reticula with metacognitive rings, the rate at which meta–ECUs/UCEs correctly flag artifacts that are later judged erroneous or fragile by external audit.

6.3 Challenge–oriented metrics

For Popper–style challenge suites:

- **Challenge pass rate:** fraction of challenge items where the ECU/UCE or reticulum behaves in accordance with the intended interpretation of M.
- **Overconfidence rate:** fraction of challenge items where the ECU/UCE or reticulum produces a determinate label instead of $\perp$ or $\perp_c$ when the domain theory supports only indeterminacy.
- **Stress stability:** behaviour of the above metrics under domain shifts or adversarial perturbations.

7 Design Patterns and Deployment Scenarios

7.1 ECU/UCE as epistemic head of an ECNN

The simplest pattern embeds a single ECU/UCE or a small multi–ECU reticulum as the epistemic head of a larger ECNN. A representation channel (Annex C) maps inputs x to representations z, and the ECU/UCE or reticulum produces artifacts a. Downstream systems and users interact only with a, not directly with z. This pattern is appropriate when:

- the domain admits a reasonably compact epistemic matrix M;
- explicit abstention and contradiction are operationally useful;
- explanations and rationales are required for audit and governance.

7.2 Multi–ECU reticula as epistemic oracles for simulators

In more complex workflows, multi–ECU reticula can act as epistemic oracles for computational simulators (e.g. multi–level physical, biological or socio–technical models). The simulator produces trajectories or states x_t; a representation channel maps x_t to z_t; the reticulum judges:

- whether trajectories are coherent with domain theory M;
- which scenarios are underexplored or epistemically fragile;
- when simulations enter regimes where the theory is silent and $\perp$ is appropriate.

This pattern is particularly natural in the RLA/CRC setting of Annex A and Annex D.

7.3 Safety–critical configurations

In safety–critical contexts (e.g. healthcare, infrastructure, finance), ECUs/UCEs and their reticula should be configured in conservative regimes:

- thresholds for accepting determinate decisions should be high;
- abstention ($\perp$) should be preferred when data or theory are insufficient;
- all ECU/UCE and reticulum versions, prompts and matrices should be versioned, logged and auditable;
- human oversight should be able to inspect artifacts and challenge decisions, updating M as needed.

The role of ECUs/UCEs and their reticula is then not to replace human judgement but to make the system's internal epistemic stance explicit and testable.

8 Discussion and Limitations

ECUs/UCEs and their reticula crystallise several trends:

- the move from raw prediction to structured epistemic outputs, including explicit ignorance and contradiction [Lipton, 2018, Miller, 2019];
- the integration of neural and symbolic components in a single, Turing–computable device [Besold et al., 2017, d'Avila Garcez et al., 2012];
- the idea that *topology* (how reasoning modules are connected) matters at least as much as raw parameter count, resonating with views in complex systems and biology [Anderson, 1972, Noble, 2012].

When ECUs/UCEs are built from LLM–based neurons, the architecture inherits both the expressive power and the potential failure modes of those models [Brown et al., 2020, Ouyang et al., 2022, Wei et al., 2022]. The epistemic matrix M and the parsing protocols become crucial choke points: they must be simple enough to be audited, yet expressive enough to encode relevant theory. The challenge is to design ECUs/UCEs and reticula that are not only powerful but also transparent and falsifiable.

This annex is deliberately abstract. Concrete instantiations will be constrained by:

- the availability and quality of domain theories encoded in M;
- the reliability, calibration and cost of LLM–based components;
- the complexity of integrating ECUs/UCEs and their reticula into existing workflows, especially in regulated domains.

Moreover, epistemic metrics are sensitive to the design of challenge suites and to assumptions about what counts as "unknown" or "contradictory". These aspects must be clarified on a per–domain basis.

9 Conclusion

Epistemic Computational Units (ECU / UCE) and their small reticula provide a concrete, formalised answer to a simple question: how can a computational system say, in a disciplined way, *this is what I think, this is why, and this is where I do not know or do not agree*?

By grounding ECUs/UCEs and their reticula in deterministic, Turing–computable maps governed by an explicit epistemic matrix M, this annex sketches a path toward architectures that:

- preserve the strengths of modern representation learning;
- make limits and contradictions first–class signals;
- support systematic testing and falsification in the spirit of Popper and computational mechanics [Popper, 1959, Shalizi & Crutchfield, 2001].

Whether ECUs/UCEs are implemented minimally or via sophisticated LLM–based neurons, their value will ultimately be measured by how well they help humans and institutions navigate complexity without hiding logical limits under a veneer of spurious certainty.

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